Fizika

9 sinf

MAVZU: Modda miqdori. Molyar massa.

Modda miqdori

Moddadagi molekula yoki atomlar sonining 12 g (${}^{12}_{6}C$) ugleroddagi atomlar soniga nisbati ***modda miqdori*** deyiladi.

ν (nyu) – 1 [mol] (XBS da asosiy birlik).

$$1\ kmol = 1000\ mol = 10^3\ mol$$

1 mol — moddaning shunday miqdoriki, undagi molekulalar soni 12 g ugleroddagi atomlar soniga teng.

Avogadro qonuni

1. Har qanday moddani 1 mol miqdorining molekulalari yoki atomlari soni teng bo‘lib u *Avogadro doimiysi deb ataladi.*

$N_A = 6{,}02 \cdot 10^{23}\ mol^{-1}$, $N = \nu \cdot N_A$

Masalan: $\nu = 2\ mol$ bo‘lsa,

$N = 2\ mol \cdot 6{,}02 \cdot 10^{23}\ mol^{-1} = 12{,}04 \cdot 10^{23}\ ta$

Avogadro qonuni

2. Har qanday gaz bir xil temperatura va bosimda bir xil hajmni egallaydi. Har qanday gazning 1 molining hajmi normal sharoitda (0℃ temperatura va 101325 Pa bosimda) $0{,}0224\ m^3 (22{,}4\ l)$ bo‘ladi va bu ***molyar hajm*** deb ataladi.

$$V_m = 0{,}0224 \frac{m^3}{mol} = 22{,}4 \frac{l}{mol} \qquad V = \nu \cdot V_m$$

Masalan, normal sharoitda 5 mol gaz hajmi:

$$V = 5\ mol \cdot 22{,}4\ l = 112\ l$$

Molyar massa

1 mol moddaning massasi *molyar massa* deb ataladi va *M* harfi bilan belgilanadi.

Birligi kg/mol yoki g/mol

$$M = m_0 \cdot N_A \Rightarrow m_0 = \frac{M}{N_A}$$

$$M_{(H_2O)} = 2 \cdot 1\frac{g}{mol} + 16\frac{g}{mol} = 18\ g/mol$$

Molekulalar soni

$$m = m_0 \cdot N \qquad m_0 = \frac{M}{N_A}$$

$$m = \frac{M}{N_A} \cdot N \Rightarrow \frac{m}{M} = \frac{N}{N_A} = \nu$$

$$N = \frac{m}{M} N_A$$

Bu formulaga ko‘ra massasi aniq bo‘lgan har qanday turdagi moddaning molekulalar (yoki atomlar) sonini aniqlash mumkin.

Molekulyar konsentratsiya

Hajm birligidagi molekulalar soniga ***molekulyar konsentratsiya*** deyiladi va $\boldsymbol{n}$ harfi bilan belgilanadi.

$$n = \frac{N}{V} \quad , \qquad [n] = m^{-3}$$

$$\frac{m}{V} = \rho$$

$$n = \frac{N}{V} = \frac{1}{V} \cdot \frac{m}{M} \cdot N_A$$

$$\frac{M}{N_A} = m_0$$

$$n = \frac{\rho}{m_0}$$

2-mashq 1-masala

Massasi 270 g bo‘lgan suvdagi modda miqdorini aniqlang.

Berilgan:

$$m = 270\ g = 0{,}27\ kg$$

$$M_{(H_2O)} = 18\,\frac{g}{mol} = 0{,}018\,\frac{kg}{mol}$$

Formula:

$$\nu = \frac{m}{M}$$

Topish kerak: $\nu - ?$

Yechish: $\nu = \dfrac{0{,}27\ kg}{0{,}018\,\frac{kg}{mol}} = 15\ mol$

Javob: $\nu = 15\ mol$

2-mashq 3-masala

Massasi 7 g bo‘lgan azot (N_2) tarkibidagi molekulalar sonini aniqlang.

Berilgan:

$m = 7\ g = 7 \cdot 10^{-3}\ kg$

$\mathrm{M}_{(\mathrm{N}_2)} = 28\,\frac{\mathrm{g}}{\mathrm{mol}} = 28 \cdot 10^{-3}\,\frac{\mathrm{kg}}{\mathrm{mol}}$

$N_A = 6 \cdot 10^{23}\ mol^{-1}$

Formula:

$N = \frac{m}{M} \cdot N_A$

Topish kerak: $N - ?$

Yechish: $N = \frac{7 \cdot 10^{-3} kg}{28 \cdot 10^{-3} \frac{kg}{mol}} \cdot 6 \cdot 10^{23} mol^{-1} = 1{,}5 \cdot 10^{23}$

Javob: $N = 1,5 \cdot 10^{23}$ **ta**

Mustaqil bajarish uchun topshiriqlar

1. 2-mashq (15-sahifa), 2-va 4-masalalarni yechish.
2. Mavzuga doir savollarga javob yozish. (15-bet).